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THIN-LAYER ELEMENT FOR INTERFACES AND JOINTS

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SUMMARY

The idea of using a thin solid element, called a thin-layer element, in soil–structure interaction and rock joints is proposed. A special constitutive model is used and various deformation modes such as no slip, slip, debonding and rebonding are incorporated. The shear stiffness is found from special laboratory tests and the normal stiffness is assumed to be composed of participation of the thin-layer element and the adjoining solid elements. A parametric study shows that the thickness of the thin-layer element can be such that the ratio of thickness to (mean) dimension of the adjacent element is in the range of 0·01 to 0·1. A number of simple and practical problems are solved to illustrate the success of the thin-layer element for soil–structure interaction problems.

INTRODUCTION

Behaviour at junctions or interfaces between structural and geological materials and at joints in rock masses involves relative translational and rotational motions under static and dynamic loading. In the context of numerical methods such as the finite element method, special interface or joint elements are used in order to account for the relative motions, and associated deformation modes.

REVIEW

One of the commonly used interface elements in soil–structure interaction is based on the joint element proposed by Goodman, Taylor and Brekke.[1] The element formulation is derived on the basis of relative nodal displacements of the solid elements surrounding the interface element (Figure 1). For two-dimensional analysis, the constitutive or stress-relative displacement relation is expressed as

$$\begin{Bmatrix} \sigma_n \\ \tau \end{Bmatrix} = \begin{bmatrix} k_n & 0 \\ 0 & k_s \end{bmatrix} \begin{Bmatrix} v_r \\ u_r \end{Bmatrix} = [C]_i \begin{Bmatrix} v_r \\ u_r \end{Bmatrix} \tag{1}$$

where σ_n = normal stress, τ = shear stress, k_n = normal stiffness, k_s = shear stiffness and v_r and u_r = relative normal and shear displacements, respectively, and $[C]_i$ = constitutive matrix for the interface or joint element. For application to soil–structure interaction problems, the thickness of the element is often assumed to be zero.

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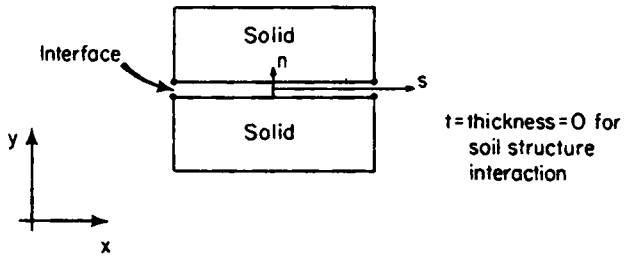


Figure 1. Relative displacement element[1]

Zienkiewicz *et al.*[2] used an isoparametric finite element formulation for an interface element which is treated essentially like a solid element. Ghaboussi, Wilson and Isenberg[3] proposed a formulation which is derived by considering relative motions between surrounding solid elements as independent degrees-of-freedom. Katona *et al.*[4,5] derived an interface model from the virtual work principle modified by appropriate constraint conditions; various deformation modes at the interface are incorporated in this formulation. Herrmann[6] presented an algorithm for an interface element which is similar to the element of Goodman *et al.*, with certain improvements through introduction of constraint conditions. He also considered various modes of interface behaviour such as sliding and debonding. Reviews and applications of interface and joint models are available in various publications.[7-11]

The foregoing elements have been incorporated into soil–structure interaction problems by various investigators. In most of these, the shear behaviour is simulated as non-linear elastic or plastic and the shear stiffness is evaluated as a tangent modulus from laboratory stress–strain behaviour in direct shear tests. Based on the assumption that the structural and geological media do not overlap at interfaces, a high value, of the order of 10^8–10^{12} units, is assigned for the normal stiffness, k_n. There is no logical basis for adoption of such values, which need to be determined for the problem on hand by performing parametric studies. Furthermore, in most problems, the formulation can provide satisfactory solutions for the stick (or no slip or bonded) and slip modes (Figure 2) for which the normal stress remains compressive. For other modes such as debonding, the solutions are often unreliable.

Development and use of a thin solid element to simulate interface behaviour is proposed herein. Since the proposed element essentially represents a solid element of small finite thickness and since it can represent a thin layer of material between two bodies, it is referred to as a 'thin-layer' element. The idea of using a thin-layer element to simulate the behaviour of interfaces appears logical and must have been attempted by many; Zienkiewicz *et al.*[2] proposed using essentially a solid element as an interface element. However, not many critical and systematic studies and implementations of the concept are available in the published literature. In recent times, a number of investigators have considered and analysed the use of the 'thin' element. The authors and their associates have been involved in research and implementation of the concept for static and dynamic problems for the last several years.[11-17] Pande and Sharma[18] compared thin and zero thickness (Figure 1) elements and studied the aspect of ill-conditioning in computations with reference to the thickness of the element. Isenberg and Vaughan[19] used a thin element in dynamic analysis.

The distinguishing features of the investigations herein lie in the special treatment of the constitutive laws for the thin-layer element, choice of its thickness, incorporation of various modes of deformation (Figure 2) and implementation for a number of problems with displacement, mixed and hybrid finite element procedures.

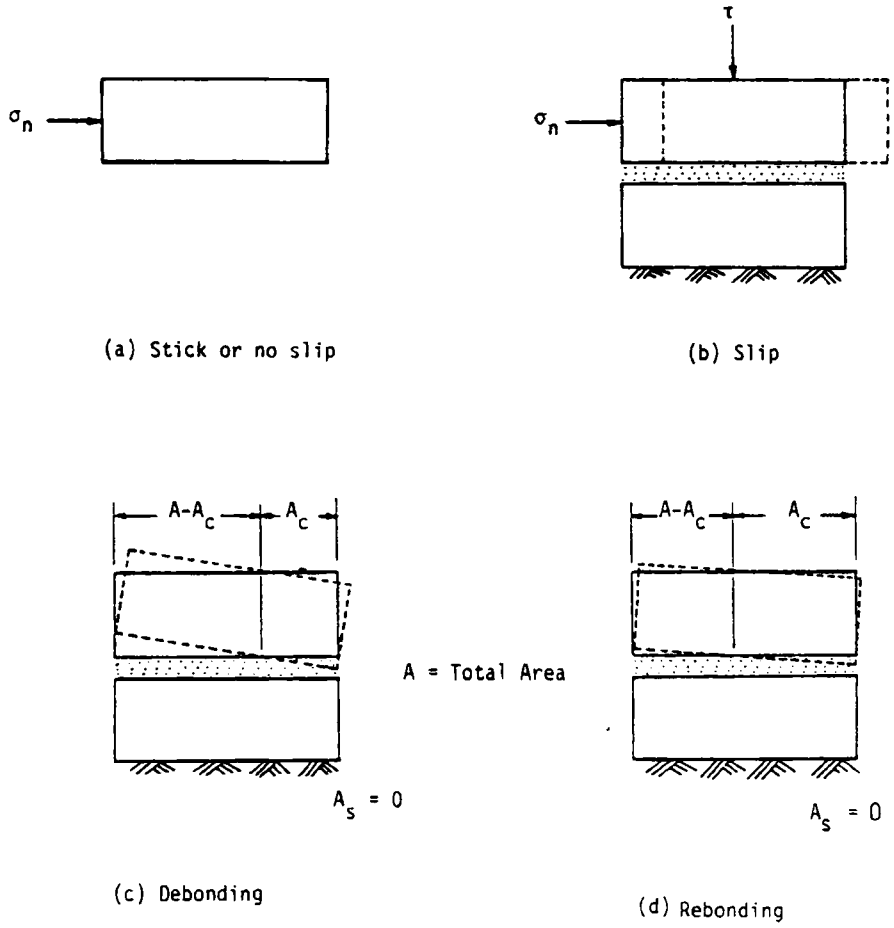


Figure 2. Schematic of modes of deformation at interface

PROPOSED THIN ELEMENT

Schematic diagrams of the thin-layer element for two- and three-dimensional idealizations are shown in Figure 3. The element is treated essentially like any other solid (soil, rock or structural) element. However, its constitutive matrix $[C]_i$ is expressed as

$$\{d\sigma\} = [C]_i\{d\varepsilon\} \tag{2a}$$

where $\{d\sigma\}$ = vector of increments of stresses, $\{d\varepsilon\}$ = vector of increments of strains, and the constitutive matrix $[C]_i$ is given by

$$[C]_i = \begin{bmatrix} [C_{nn}]_i & [C_{ns}]_i \\ [C_{sn}]_i & [C_{ss}]_i \end{bmatrix} \tag{2b}$$

where $[C_{nn}]$ = normal component, $[C_{ss}]$ = shear component and $[C_{ns}]$, $[C_{sn}]$ represent coupling effects. Since it is difficult to determine the coupling terms from laboratory tests, at this time, they are not included.

A basic assumption made is that the behaviour near the interface involves a finite thin zone (Figure 3) rather than a zero thickness as assumed in previous formulations. As stated earlier,

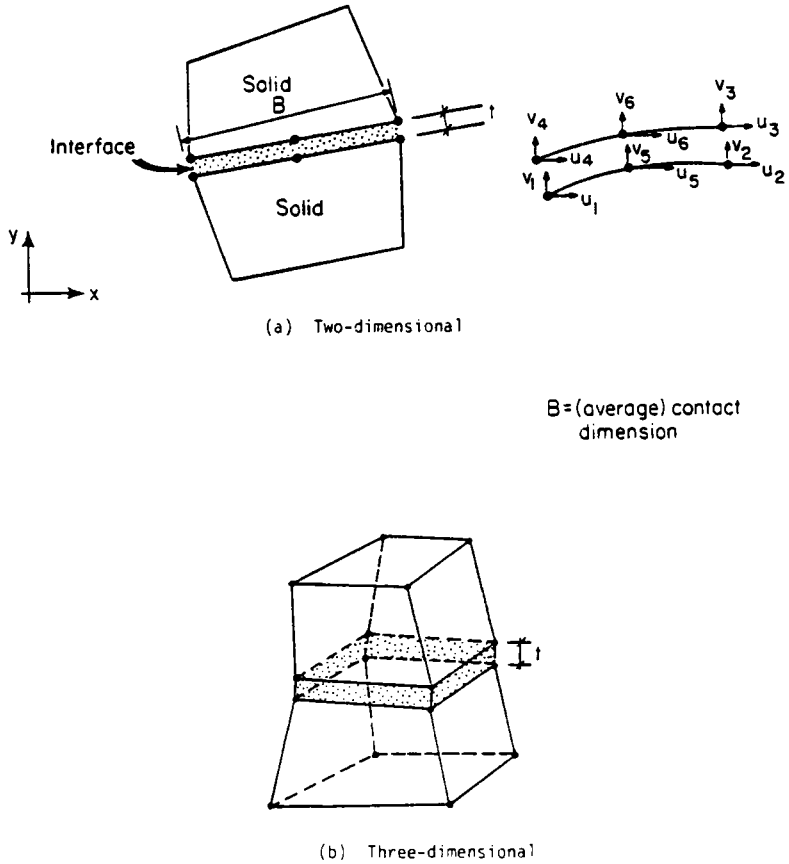


Figure 3. Thin-layer interface element

it may not be appropriate to assign an arbitrarily high value for the normal stiffness. Since the interface is surrounded by the structural and geological materials, its normal properties during the deformation process must be dependent upon the characteristics of the thin interface zone as well as the state of stress and properties of the surrounding elements. Based on these considerations, it was proposed to express the normal stiffness as[11]

$$[C_n]_i = [C_n(\alpha^i_m, \beta^g_m, \gamma^{st}_m)] \tag{3}$$

where $\alpha^i_m, \beta^g_m, \gamma^{st}_m$ $(m = 1, 2, \ldots)$ denote the properties of the interface, geological and structural elements, respectively. From here onwards the second subscript in $[C_{nn}]$ etc. is dropped. Equation (3) can be written as

$$[\bar{C}_n]_i = \lambda_1[C_n]_i + \lambda_2[C^g_n] + \lambda_3[C^{st}_n] \tag{4}$$

where $[\bar{C}_n]_i$ denotes normal behaviour of the thin interface element, and λ_1, λ_2 and λ_3 are the participation factors varying from 0 to 1. Equation (4) is expressed as an addition of various components. However, it is possible to define it by using a different (polynomial) expression. One of the simplifications would be to assume $\lambda_2 = \lambda_3 = 0$ and $\lambda_1 = 1$, implying that the normal component is based on the normal behaviour of the thin-layer element evaluated just as the adjacent soil element. It is possible to arrive at appropriate values for λ_1 based on a trial and

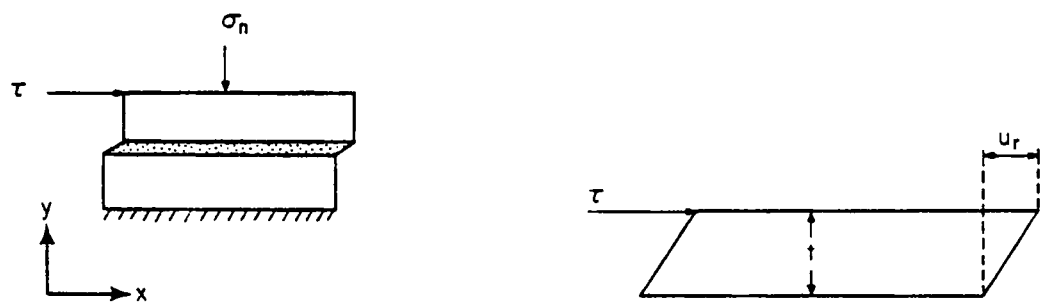
error procedure in which numerical solutions are compared with laboratory or field observations. Often, it was found that satisfactory results can be obtained by assigning the interface normal component the same properties as the geological material.[12–16]

For most applications presented herein, $\lambda_1 = 1$ and $\lambda_2 = \lambda_3 = 0$ were adopted. As long as the significant deformation mode is stick, these values provide satisfactory results. The contribution of the participation factors becomes important when opening or debonding initiates. Preliminary work towards determination of the participation factors for normal stiffness in cyclic loading has shown $\lambda_1 = 0{\cdot}75$, $\lambda_2 = 0{\cdot}25$ and $\lambda_3 = 0{\cdot}00$.[16,17] Now, work towards derivation of these factors by comparison with laboratory data is in progress.

The shear component $[C_s]_i$ is obtained from direct shear or other interface shear testing devices (Figure 4(a)). In this study $[C_s]_i$ is assumed to be composed of a shear modulus G_i for the interface. The expression used for tangent G_i is given by

$$G_i(\sigma_n, \tau, u_r) = \frac{\partial[\tau(\sigma_r, u_r)]}{\partial u_r} \times t \bigg|_{\sigma_n} \tag{5}$$

where t = thickness of the element (Figure 4(b)) and u_r = relative displacement.



(a) Schematic of Direct Shear Test (b) Deformations at the Interface

Figure 4. Behaviour at interface

Constitutive modelling

The thin-layer interface element can be formulated by assuming it to be linear elastic, non-linear elastic or elastic-plastic. The development of its stiffness characteristic follows essentially the same procedure as solid elements, that is the stiffness matrix, $[k]_i$ is written as

$$[k]_i = \int_V [B]^T [C^{ep}]_i [B] \, dV \tag{6a}$$

where $[B]$ = transformation matrix, V = volume and $[C^{ep}]_i$ is the constitutive matrix. Then the element equations are written as

$$[K]_i \{q\} = \{Q\} \tag{6b}$$

where $\{q\}$ = vector of nodal displacements and $\{Q\}$ = vector of nodal forces.

For linear elastic behaviour, $[C^e]_i$ can be expressed as

$$[C^e]_i = \begin{bmatrix} C_1 & C_2 & C_2 & 0 & 0 & 0 \\ C_2 & C_1 & C_2 & 0 & 0 & 0 \\ C_2 & C_2 & C_1 & 0 & 0 & 0 \\ 0 & 0 & 0 & G_{i1} & 0 & 0 \\ 0 & 0 & 0 & 0 & G_{i2} & 0 \\ 0 & 0 & 0 & 0 & 0 & G_{i2} \end{bmatrix} = \begin{bmatrix} [C_n]_i & 0 \\ 0 & [C_s]_i \end{bmatrix} \tag{7}$$

where

$$C_1 = \frac{E(1-\nu)}{(1+\nu)(1-2\nu)}$$

$$C_2 = \frac{E\nu}{(1+\nu)(1-2\nu)}$$

E is the elastic (Young's) modulus, ν is Poisson's ratio and G_{ii} ($i = 1, 2, 3$) are the shear moduli defined in equation (5). If the shear behaviour is assumed to be isotropic, $G_{i1} = G_{i2} = G_{i3}$. Here it is assumed that the shear response is uncoupled from the normal response represented by $[C_n]$. For two-dimensional plane strain idealization, the special form of $[C^e]_i$ and its inverse form, $[D^e]_i$ are given as

$$[C^e]_i = \begin{bmatrix} C_1 & C_2 & 0 \\ C_2 & C_1 & 0 \\ 0 & 0 & G_i \end{bmatrix} \tag{8}$$

$$[D^e]_i = \begin{bmatrix} \dfrac{1-\nu^2}{E} & \dfrac{-\nu(1+\nu)}{E} & 0 \\ \dfrac{-\nu(1+\nu)}{E} & \dfrac{1-\nu^2}{E} & 0 \\ 0 & 0 & \dfrac{1}{G_i} \end{bmatrix} \tag{9}$$

the latter is used in a mixed finite element procedure.[13,20]

For non-linear elastic behaviour such as hyperbolic simulation, E, ν and G can be defined as variable moduli based on triaxial and direct shear tests.[21–23] For example G_i can be written as

$$G_i = K\gamma_w \left(\frac{\sigma_n}{p_a}\right)^n \left(1 - \frac{R_{f\tau}}{c_a + \sigma_n \tan\phi}\right)^2 \tag{10}$$

where K, n and R_f are material parameters, γ_w = unit weight of water, p_a = atmospheric pressure, c_a = cohesion and ϕ = angle of friction. Alternatively, the shear modulus can be obtained by expressing the τ vs. u_r relation in a polynomial form described subsequently.

In addition to the foregoing linear and non-linear elastic models, the proposed formulation also allows for elastic–plastic behaviour. Then the constitutive matrix for the interface is written as

$$[C]_i^{ep} = [C^e(k_s, k_n)]_i - [C^p(k_s, k_n, \{du_r^p\})]_i \tag{11}$$

where $\{du_r^p\}$ = vector of incremental relative displacements. The second part of equation (11) is found on the basis of yield and flow criteria of the theory of plasticity. Here conventional criteria such as Mohr–Coulomb can be used with yield function, f, and plastic potential function, Q; for associated plasticity $f = Q$. To allow for dilatancy in the case of rock joints different f and Q can be used in the context of non-associative plasticity.

Stick, slip, debonding and rebonding modes

Depending upon the material model used, criteria such as Mohr–Coulomb and Drucker–Prager are used to define the initiation of slip at the interface; before the slip, the interface is assumed to be in the stick or no slip mode (Figure 2(a)). The stick mode and the slip mode can occur such that the normal stress is compressive. During debonding, the computed normal stress can be tensile, but physically, it is zero at the interface. In subsequent loading or unloading, an interface that has experienced opening or debonding may close or rebond. This is identified when the normal stress in the interface becomes compressive. In soil–structure interaction problems, penetration of nodes is not permitted during closing. In a non-linear incremental analysis, the (excess) tensile stress is redistributed in the surrounding zones during the iterations at each load increment. Thus the stick, slip, debonding and rebonding are accounted for in the proposed procedure as discussed in Reference 11. A variational formulation for development of appropriate element equations to account for various modes is described in Reference 16. Details of the procedure for handling various deformation modes in Figure 2 and for controlling penetration of nodes are given in Appendix I.

Testing for interface modelling

Reviews of test devices relevant for laboratory determination of the constitutive parameters are presented in References 7 and 11. For the static and cyclic behaviour of interfaces, a new piece of equipment called a cyclic multi-degree-of-freedom device has been developed.[24] It can permit testing for interfaces and joints under stress and strain controlled loading and for translational, torsional and rocking modes.

CHOICE OF ELEMENT THICKNESS

The quality of simulation of the interface behaviour will depend on a number of factors such as physical and geometrical properties of the surrounding media, non-linear material behaviour and the thickness of the thin-layer element. If the thickness is too large in comparison with the dimension B, of the surrounding elements (Figure 3), the thin-layer element will behave essentially as a solid element. If it is too small, computational difficulties may arise. The choice of thickness can, therefore, be an important question, and can be resolved by performing parametric studies in which the predictions from various thicknesses are compared with observations. The choice of thickness can become particularly important for dynamic analysis where the mass and damping properties need to be considered. A preliminary study performed to achieve this goal is described in the following.

Figure 5(a) shows a schematic diagram of a direct shear test device in which the bottom half is concrete and the top half contains soil. A series of tests were performed with concrete–sand interface under different normal loads.[22,23] Figure 6 shows two typical test results for the sand with a relative density around 80 per cent for two normal loads.

The direct shear test data for the normal stress of $\sigma_n = 4{\cdot}89$ t/ft^2 (4·77 kg/cm^2) and $\sigma_n = 9{\cdot}79$ t/ft^2 (9·55 kg/cm^2) were analysed by using a two-dimensional plane strain finite element

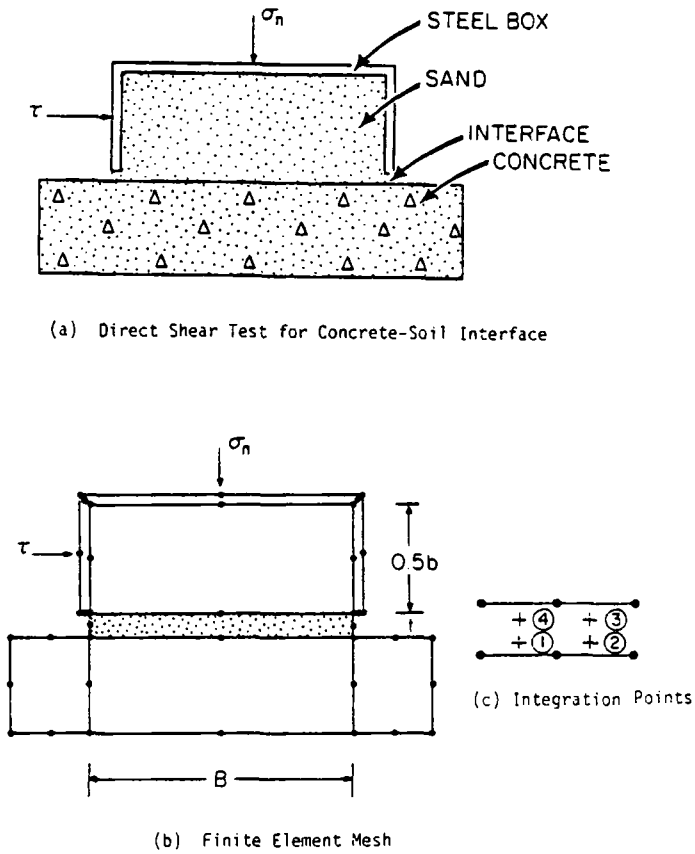


Figure 5. Analysis for direct shear test for choice of thickness

procedure. The shear stress vs. relative displacement curve was simulated by using a polynomial function:

$$\tau = \alpha_1 u_r + \alpha_2 \sigma_n u_r + \alpha_3 u_r^2 + \alpha_4 \sigma_n^2 u_r^2 \tag{12}$$

where α_1 ($i = 1, 2, \ldots, 4$) are constants. Here the values of α_i were found to be $\alpha_1 = 53$ kg/cm^3, $\alpha_2 = 8{\cdot}2$ cm^{-1}, $\alpha_3 = 303$ kg/cm^4, $\alpha_4 = 7{\cdot}0$ kg^{-1}. The tangent stiffness k_s in equation (5) was evaluated as the derivative of τ with respect to u_r at a point during the incremental loading.

The finite element mesh using eight-noded isoparametric elements is shown in Figure 5(b). The outside metallic box was also discretized and the shear load was applied at the mid-node on its vertical side. The material properties used are given below:

Soil

Initial modulus: $E_s = 1{,}000$ kg/cm^2

Poisson's ratio: $\nu_s = 0{\cdot}375$

Concrete

$E_c = 1{\cdot}5 \times 10^5$ kg/cm^2

$\nu_c = 0{\cdot}2$

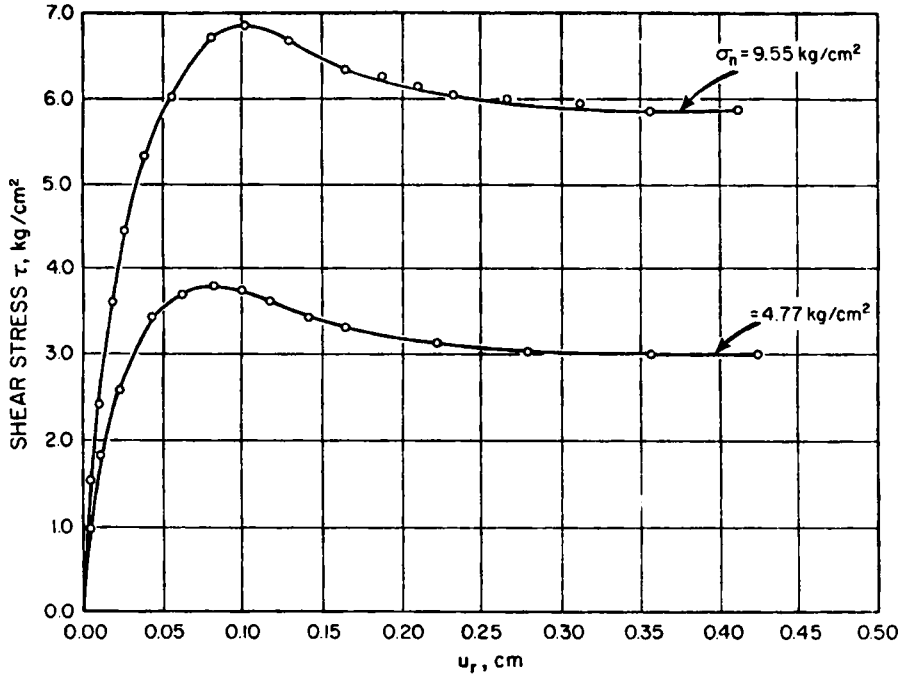


Figure 6. Typical direct shear test results, $D_r = 80$ per cent[23]

Interface

Adhesion,	$c_a = 0{\cdot}0$
Angle of friction,	$\phi = 38°$, $\tan\phi = 0{\cdot}7813$
Initial modulus,	$E_i = 10^3$ kg/cm^2
	$\nu_i = 0{\cdot}375$

The thickness, t, of the element was varied such that the ratio t/B was 0·001, 0·010, 0·100 and 1·00. Since the shear modulus G_i is a function of the thickness, its value also varied with the thickness.

The computed results in terms of (relative) displacement, u_r, and the ratio t/B for applied shear stress $\tau = 3{\cdot}0$ kg/cm^2, $\sigma_n = 4{\cdot}77$ kg/cm^2, and $\tau = 5{\cdot}00$ kg/cm^2 for $\sigma_n = 9{\cdot}55$ kg/cm^2 are plotted in Figure 7. The computed displacements show a wide variation as the thickness is changed. However, the results in the range of 0·01 to 0·1 of t/B show satisfactory agreement with the observed values of 0·032 cm and 0·034 cm for the tests with $\sigma_n = 4{\cdot}77$ kg/cm^2 and 9·55 kg/cm^2, respectively (Figure 6). Note that for higher and lower values of t/B than this range, the computed displacements are much different from the measured values.

Table I shows computed values of shear stresses at the four integration points (Figure 5(c)) of the interface element. It can be seen that the best correlation between computations and the applied stresses of $\tau = 3{\cdot}00$ kg/cm^2 and 5·00 kg/cm^2 is obtained for a t/B ratio in the range of 0·001 to 0·1. Although the average values of the computed stresses are not significantly different for the ratios considered, the uniformity of computed stresses is greater for t/B ratios smaller than 0·1.

Based on the foregoing results, it can be concluded that satisfactory simulation of interface behaviour can be obtained for t/B ratios in the range from 0·01 to 0·1. This conclusion may

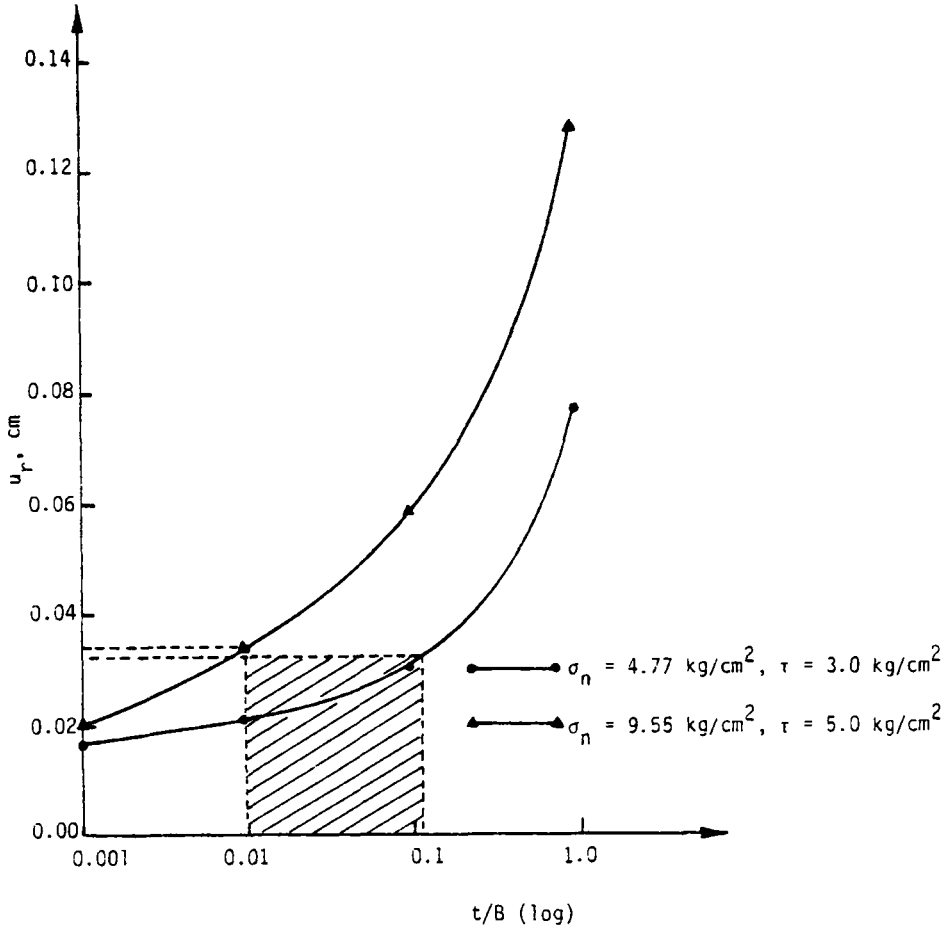


Figure 7. Effect of thickness of interface element

Table I. Distribution of shear stress in interface element at integration points (Figure 5(c))

$\sigma_n = 4{\cdot}77$ kg/cm^2	Integration points				
t/B	1	2	3	4	Average
1	2·3939	3·6034	2·1203	3·8771	2·9987
0·1	2·7390	3·2602	2·7049	3·2940	2·9996
0·01	3·0022	2·9975	2·9975	3·0022	2·9999
0·001	2·9967	3·0021	2·9964	3·0025	2·9994
$\sigma_n = 9{\cdot}55$ kg/cm^2					
1	3·7868	6·2084	3·239	6·7562	4·9976
0·1	4·4778	5·5210	4·4093	5·5895	4·9994
0·01	5·0044	4·9950	4·9950	5·0044	4·9997
0·001	4·9914	5·0065	4·9905	5·0075	4·9990

need (small) modifications if the non-linear behaviour of soils and interfaces were simulated by using other or advanced constitutive models.

Pande and Sharma[18] reported that thin element provided satisfactory results for much lower ratios of t/B. From this study, however, it is felt that from the viewpoints of both the displacements and stresses, use of the ratio t/B in the range 0·01 to 0·1 can be appropriate. The ratio can vary during mesh refinement; however, if it remains within this range, the thin-layer element can provide satisfactory simulation of the interface behaviour.

APPLICATIONS

The proposed thin-layer interface element has been introduced in displacement, mixed and hybrid finite element methods,[12–16,20] in the case of the displacement method, both two- and three-dimensional formulations have been developed. This paper describes a number of simple and practical applications in static soil–structure interaction. The concept is also incorporated in a dynamic finite element analysis.[11,16,17]

Example 1: single element, linear elastic material

Figure 8 shows a single interface element idealized as plane strain with length equal to unity, and $t/B = 0{\cdot}01$. A surface shear traction, T_x, equal to 10 units is applied at the top of the element. The linear elastic material properties assumed are

$$E = 1{,}000 \text{ units}$$

$$\nu = 0{\cdot}3$$

$$G_i = 10 \text{ units}$$

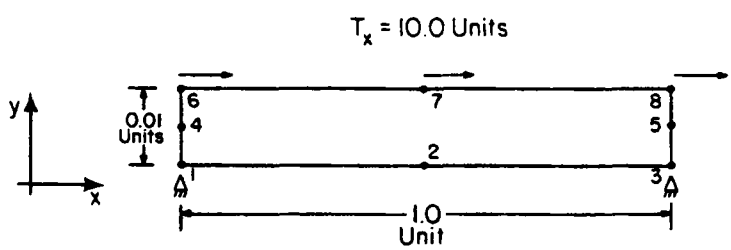


Figure 8. Interface tests for linear behaviour

Table II shows computed results for the three stress components and two displacement components. The computed shear stress is very close to the applied shear stress. The computed displacements of about 10^{-2} units at the top modes are close to the exact solution given by

$$\delta = \frac{\tau t}{G_i} = \frac{10 \times 0{\cdot}01}{10} = 0{\cdot}01 \text{ units} \tag{13}$$

Example 2: two-dimensional case: linear elastic behaviour

Figure 9(a) shows two plane strain elements with a thin interface element. The elements are loaded with different vertical tractions. The properties of the solid and interface elements are

Table II. Results for interface test: example 1

Node	σ_x	σ_y	τ_{xy}	x-displacement $\times 10^{-2}$	y-displacement $\times 10^{-6}$
1	0·25714	0·60000	10·0035	0·0	0·0
2	0·0	0·0	9·99825	0·0	0·0
3	−0·25714	−0·60000	10·0035	0·0	0·0
4	0·11699	0·30000	10·0035	0·500177	1·68272
5	−0·11699	−0·30000	10·0035	0·500177	−1·68272
6	−0·00232	0·0	10·0035	1·00035	2·27373
7	0·0	0·0	9·99825	0·99827	0·0
8	0·02316	0·0	10·0035	1·00035	−2.27373

Solid

$E = 10{,}000$ psi (69×10^3 kPa)
$\nu = 0{\cdot}3$

Interface

$E = 1{,}000$ psi (69×10^3 kPa)
$\nu = 0{\cdot}30$
$G_i = 20$ psi (138 kPa)
$t = 0{\cdot}1 = (0{\cdot}01B)$ inch (0·254 cm)

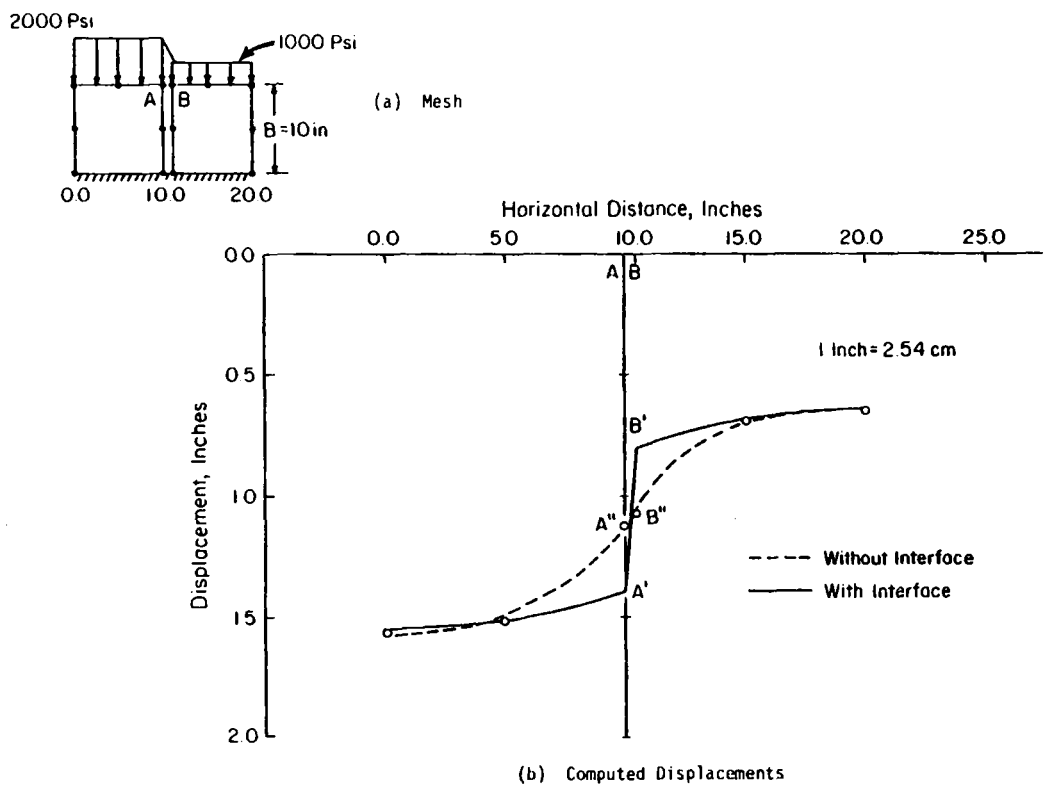


Figure 9. Two-dimensional test

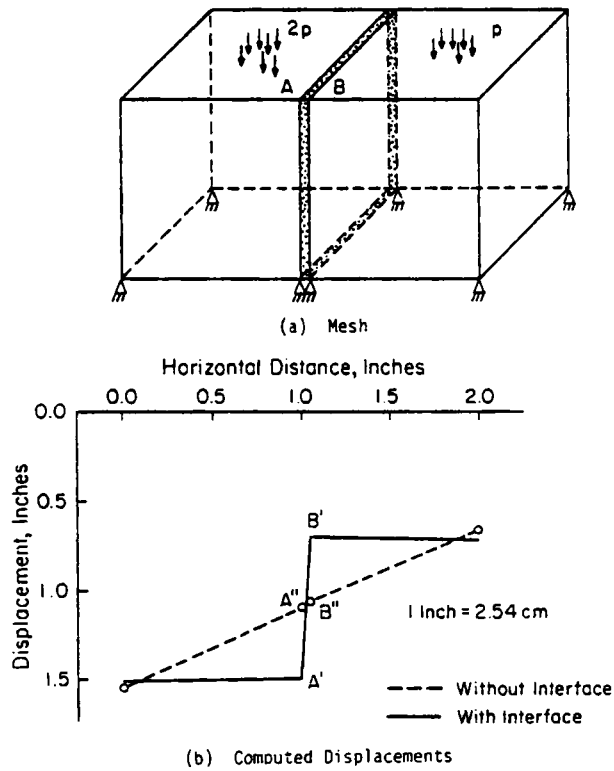


Figure 10. Three-dimensional test

The results for vertical displacements of two typical points A and B with and without interface are shown in Figure 9(b). It can be seen that provision of the interface shows significant relative movements between the two solid elements, which appear to be satisfactory.

Example 3: three-dimensional case: linear elastic behaviour

The finite element mesh is shown in Figure 10(a). The loading and the material properties are similar to those in Example 2. The computed displacements at points A and B (Figure 10(b)) are similar to those in the two-dimensional case. As before, the provision of the interface shows significant relative motions as compared to those without the interface element.

Example 4: buried pipe

Figure 11 shows an idealized pipe–soil system analysed by Katona.[5] The soil medium is assumed to be homogeneous and linearly elastic, and the pipe material is also assumed to be linear elastic. Three cases of interface friction are considered: $\tan\phi = 0{\cdot}001$, $\tan\phi = 2{\cdot}0$ and $\tan\phi = 0{\cdot}25$ to represent approximately frictionless slip, stick (bonded or no slip case), and frictional slip, respectively. The value of $\tan\phi$ is used to define slip according to the following Mohr–Coulomb criterion:

$$\begin{aligned} \tau &\geqslant c_a + \sigma_n \tan\phi \\ &\geqslant S \end{aligned} \tag{14}$$

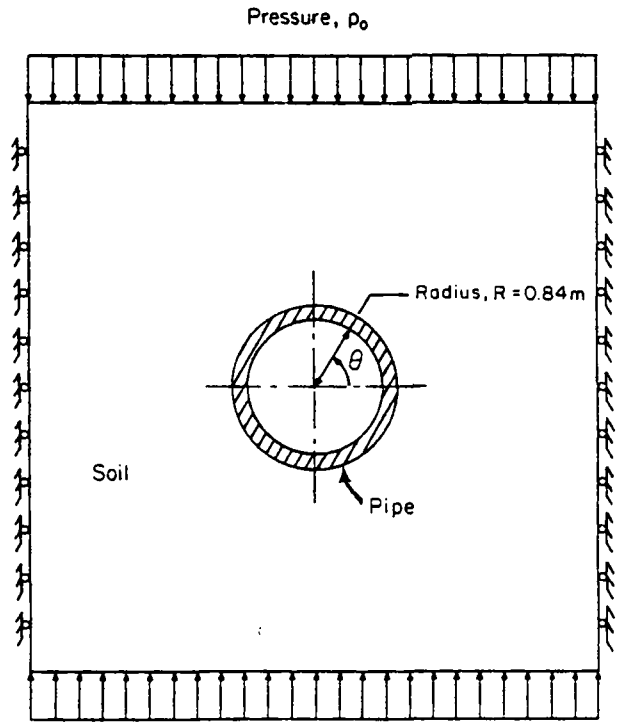


Figure 11. Buried pipe

where c_a = adhesion assumed to be zero, σ_n = normal stress, S = shear strength and τ = induced shear stress. The other material properties assumed are

Soil

$E_s = 3{,}000$ psi (20,670 kPa)
$\nu_s = 0{\cdot}333$

Pipe

$E_p = 2{\cdot}2 \times 10^6$ psi ($15{\cdot}2 \times 10^6$ kPa)
$\nu_p = 0{\cdot}25$

Interface

$G_i = 128{\cdot}8$ t lb/in^2 ($3{\cdot}56$ t $\times 10^6$ kPa)
$\nu_i = 0{\cdot}333$
$\tau = 128{\cdot}8\,(u_r) - 250\,u_r^2$
$t = 0{\cdot}01\,(\pi R/8)$

where R = radius of the pipe = 0·84 m. The deformation behaviour of the interface is assumed to be the same for all the three cases. It may be noted that in the closed-form solution used subsequently, the interface behaviour is expressed through relative properties of soil and structure, and the deformation properties of the interface itself do not enter directly into the calculations.

The finite element mesh is shown in Figure 12. The soil is assumed to be weightless and the overburden pressure p_0 is applied in a number of increments.

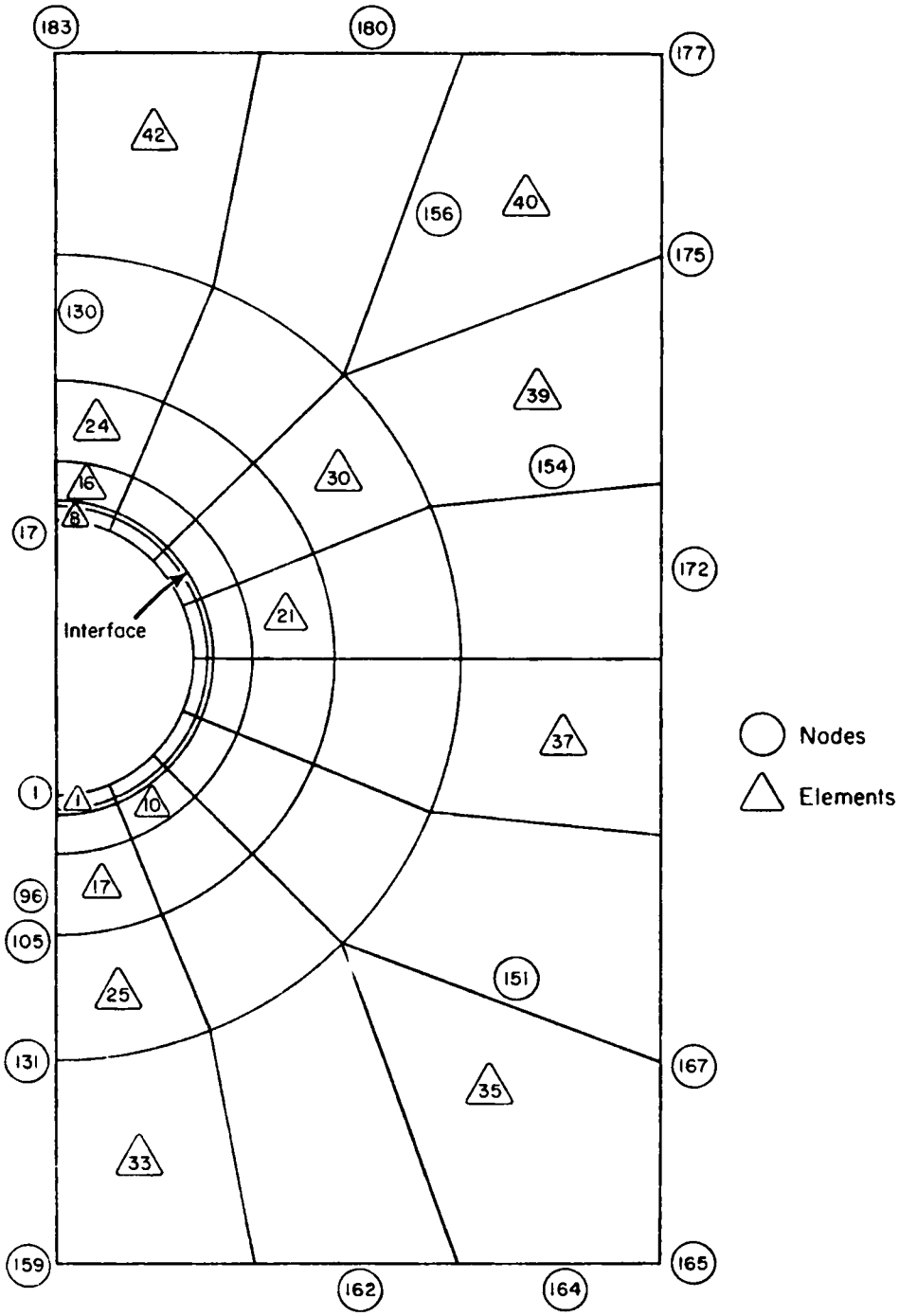
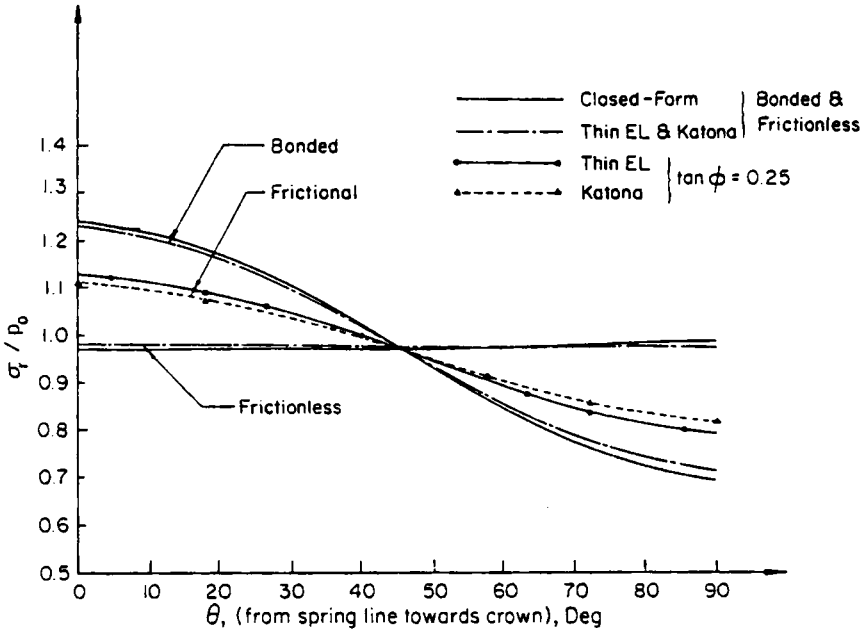
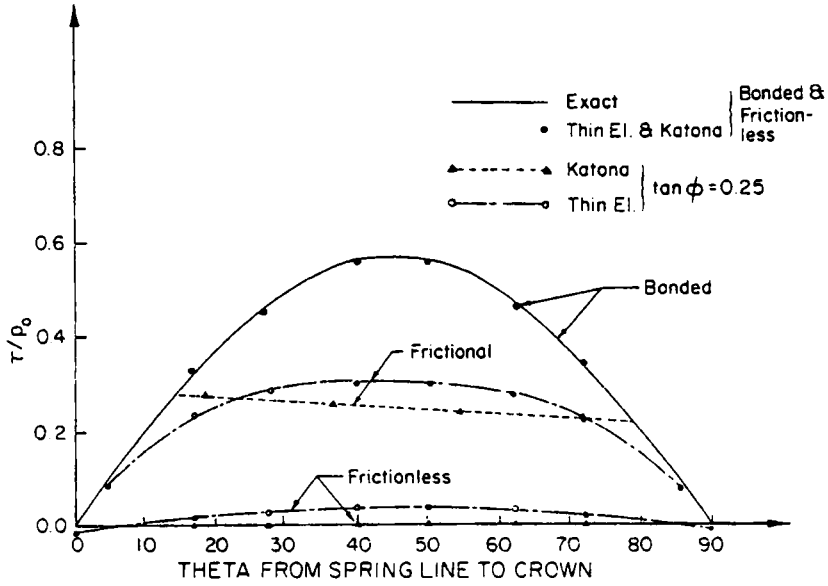


Figure 12. Mesh for buried pipe

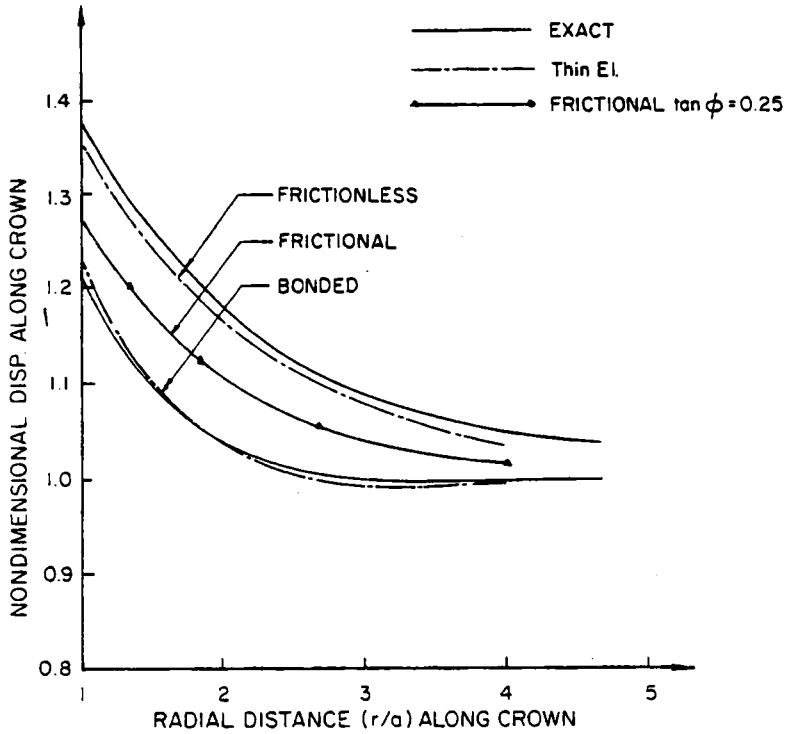


(a) σ_r/p_0 vs θ



(b) τ/p_0 vs θ

Figure 13. Comparison of numerical and closed-form solutions



(c) Displacements along crown

Figure 13. (Continued)

Figure 13(a) shows comparison for ratio of radial stress, σ_r, to p_0 vs. θ (from springline to crown) between the predictions for the three cases from the present analysis, Katona's results and exact solutions. The exact solution is obtained from a closed-form elasticity procedure developed by Burns and Richards,[25] which is available only for the frictionless and bonded cases. Figure 13(b) shows similar comparisons for τ/p_0 vs. θ. The present computations compare closely with the exact solutions and the predictions by Katona[5] which are based on a formulation using the virtual work principle and special constraint conditions. The results for τ/p_0 with $\tan\phi = 0{\cdot}25$ show non-linear variation whereas Katona's results show linear variation. For the frictionless case, the present results show small non-zero values for τ/p_0; this can be because the frictionless case is simulated by adopting a small value of $\tan\phi$ $(= 0{\cdot}001)$.

Figure 13(c) shows comparisons between the present predictions for displacement along the crown and the closed-form solutions. Katona's paper[5] did not include these predictions, hence, they do not appear in Figure 13(c). Again, the correlation between the present procedure and the exact solutions for displacements is satisfactory.

Example 6: retaining wall

Mixed and hybrid finite element schemes with the thin-layer interface element were used to predict the behaviour of a strutted retaining wall (Figure 14).[26] Numerical predictions from these procedures were compared with field observations; typical results for only the mixed procedure are included herein.

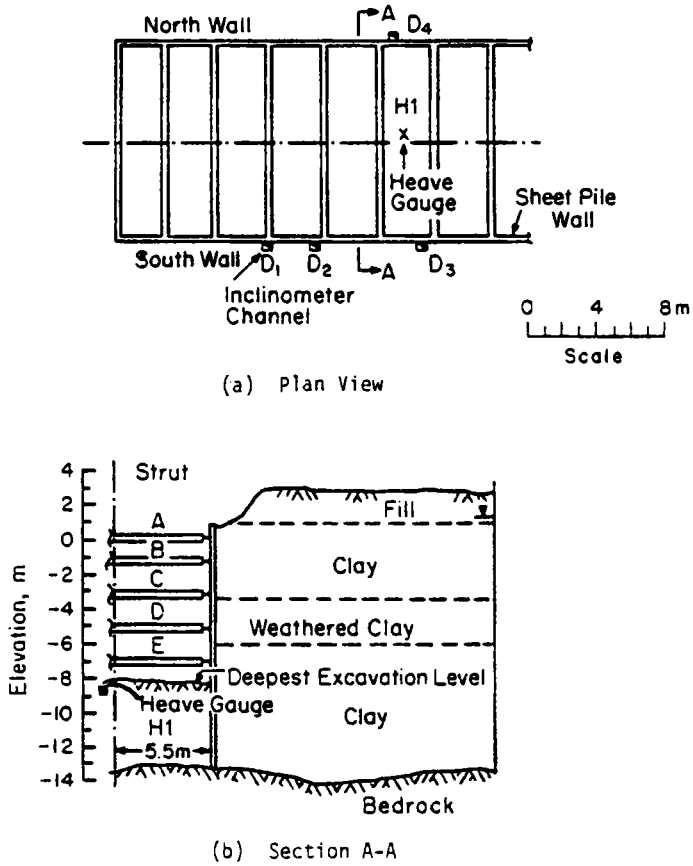


Figure 14. Details of retaining wall[26]

Table III. Soil properties for strutted wall[13,26,27]

Material number	E (t/m^2)	ν	c (t/m^2)	ϕ	γ (t/m^3)	K_0
1	800·0	0·4933	4·0	0	1·95	0·65
2	760·0	0·4933	3·8	0	1·95	0·65
3	900·0	0·4933	4·5	0	1·95	0·65
4	500·0	0·4967	5·0	0	1·95	0·65
5	410·0	0·4967	4·1	0	1·95	0·65
6	368·0	0·3000	1·42	0	1·95	0·65

Wall properties $E = 2{\cdot}76 \times 10^6$ t/m^2, $\nu = 0{\cdot}33$

Interface properties $E = 450{\cdot}0$ t/m^2, $\nu = 0{\cdot}33$, $G_i = 0{\cdot}10$ t/m^2, t = 0·05 m

Strut properties

A, D: Area = 1·0 m^2, $E = 912{\cdot}0$ t/m^2

1t = 1000 kg

B, C, E: Area = 1·0 m^2, $E = 4560{\cdot}0$ t/m^2

Table IV. Sequences of construction[26]

Stage	Detail
1	*In-situ* stresses, install wall, excavate to +2·0 m
2	Install strut A, excavate to −2·0 m
3	Install strut B, excavate to −3·0 m
4	Install strut C, excavate to −4·0 m
5	Excavate to −5·0 m
6	Install strut D, excavate to −6·0 m
7	Excavate to −7·0 m
8	Install strut E, excavate to −8·0 m

The material properties used are shown in Table III; various construction sequences simulated are shown in Table IV. The wall and strut were assumed to be linearly elastic, the soil was treated as elastic–plastic material obeying von Mises rule and the interface was assumed to be non-linear elastic with slip defined by the Mohr–Coulomb criterion. Figure 15 shows the finite element mesh. The t/B ratio used was about 0·01.

Since the main objective here is to show the usefulness of the thin interface element in computing the stresses at and near the interfaces, the results presented herein include comparison of wall pressures for various stages of the construction sequence. Figures 16(a) to (d) show comparisons between the results of a mixed procedure using the thin interface, and the field observations for wall pressures after 2, 3, 5 and 7 stages of the construction stages. In

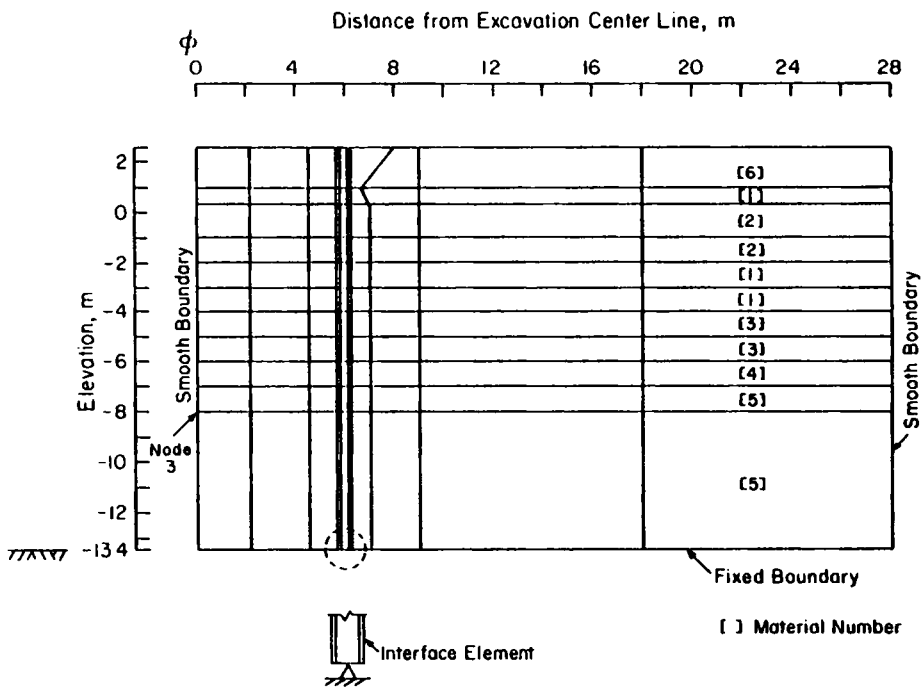


Figure 15. Finite element mesh for Vaterland 1 wall[13,26,27]

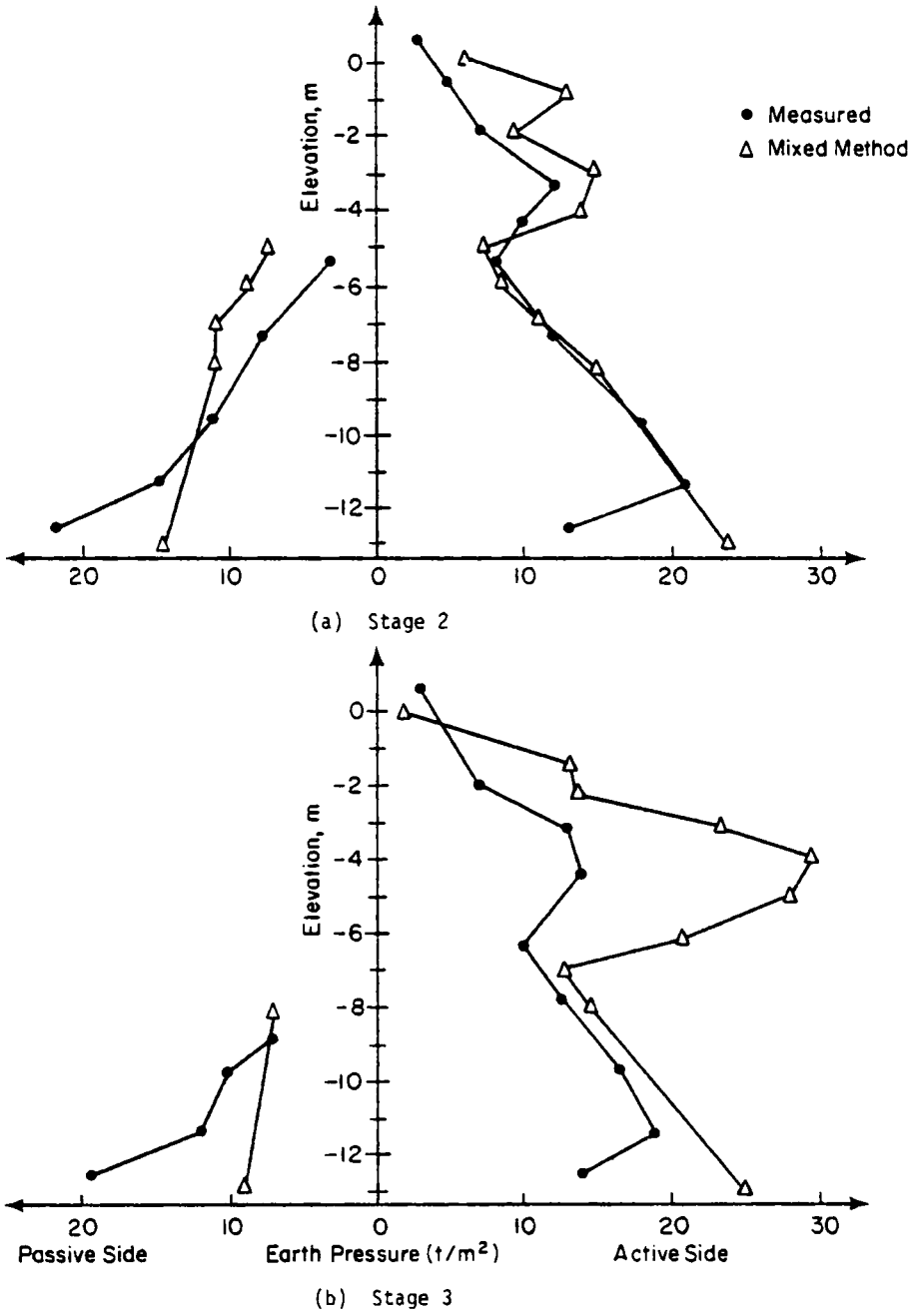
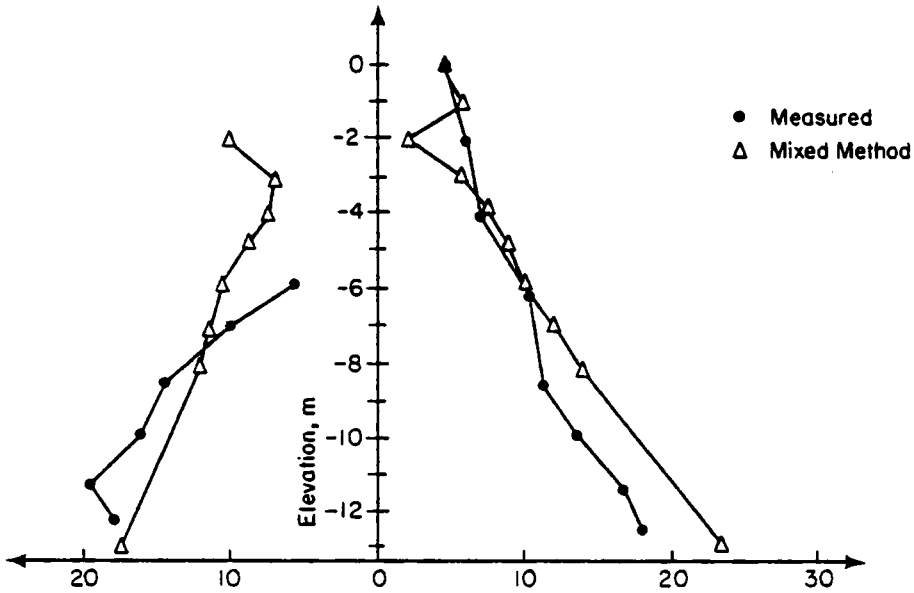
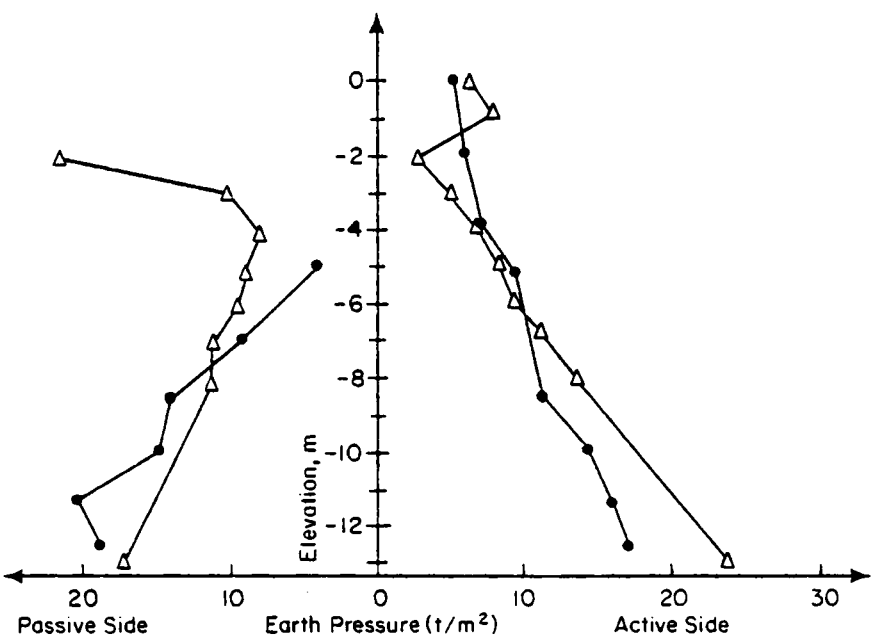


Figure 16. Comparisons of predicted and observed wall pressures



(c) Stage 5



(d) Stage 7

Figure 16. (Continued)

view of the uncertainties inherent in the material parameters, field tests and measurements, the correlation between the predictions and observations is considered to be satisfactory.

One of the major improvements provided by the thin-layer element lies in its capability to provide consistent and satisfactory computations of stresses in the interfaces themselves, whereas with the zero thickness element, it has been often found difficult to obtain consistent and stable stresses in the interface elements. As a result, most past studies have plotted wall pressures in the adjoining soil elements rather than in the interfaces.[9,21,27,28]

CONCLUSIONS

The concept of thin-layer elements to simulate the behaviour at junctions between dissimilar media such as interfaces in soil–structure interaction and rock joints is used herein. The constitutive law for the thin interface element is defined by expressing the constitutive matrix in terms of normal and shear characteristics. The former is related to adjoining elements, and the latter is defined from laboratory tests. Consideration is given to deformation modes such as stick, slip, debonding and rebonding and to the choice of appropriate thickness. A number of example problems are solved and the predictions with displacement, mixed and hybrid procedures are compared with closed-form solutions and field observations.

Based on the investigations herein, the following conclusions are made:

(1) The thin-layer element can provide satisfactory solutions for many two- and three-dimensional problems.
(2) It is capable of providing improved definition of normal and shear behaviour, hence, it can be computationally more reliable than the zero thickness element.
(3) With appropriate incorporation of stress redistribution schemes, it is possible to handle various deformation modes such as stick, slip, debonding and rebonding.
(4) Since its formulation is essentially the same as other solid elements, it is easier to programme and implement.
(5) Inclusion of a finite thickness for the interface can be realistic since there is very often a thin layer of soil which participates in the interaction behaviour. As a result, remoulding and degradation in the interface zones between structures and geological media can be incorporated.
(6) Although this study has concentrated on applications in soil–structure interaction, the concept can be applicable for joints in rocks.
(7) In view of the success in the use of the simple thin element, it may not be necessary to develop more complex approaches unless they are required by particular problem(s).

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APPENDIX I

(A) Modes of deformation

Computed	After iteration and equilibrium
Stick mode (Figure 2(a))	
$\sigma_n > 0$, $\tau < S$, S = Shear strength	Same as computed
Slip mode (Figure 2(b))	
$\sigma_n > 0$, $\tau \geq S$	$\sigma_n^1 > 0$, $\tau^1 = S$; Superscript ^{1} denotes value after iteration
Debonding (Figure 2(c))	
$\left.\begin{matrix}\sigma_n > 0\\ \tau < S\end{matrix}\right\}$ on A_c	$\left.\begin{matrix}\sigma_n^1 > 0\\ \tau^1 < S\end{matrix}\right\}$ on A_c^1
$\left.\begin{matrix}\sigma_n > 0\\ \tau \geq S\end{matrix}\right\}$ on A_s	$\left.\begin{matrix}\sigma_n^1 > 0\\ \tau^1 = S\end{matrix}\right\}$ on A_s^1
$\left.\begin{matrix}\sigma_n \leq 0\\ \tau \geq 0\end{matrix}\right\}$ on A-A_c-A_s	$\left.\begin{matrix}\sigma_n^1 = 0\\ \tau^1 = 0\end{matrix}\right\}$ on A-A_c^1-A_s^1

A = total area, A_c = contact area, A_s = area of slip.
A_s is usually assumed to be zero.

Rebonding (Figure 2(d))	
$\sigma_n > 0$	$\sigma_n^1 > 0$
$\tau \geq 0$	$\tau^1 \geq 0$

As an approximation, at this time, full contact is assumed on rebonding.

(B) Check for interpenetration of nodes

Figure 17 shows a schematic diagram of a thin-layer element with two typical adjacent nodes p and q. With the thin-layer formulation, generally the nodes do not penetrate each other. However, if penetration occurs, it is prevented by separating them by an (arbitrary) fraction (ε) of the thickness t. The steps in the algorithm to achieve this are given below.

(1) Compute normal displacements, u_n^i, referred to the local co-ordinate system (s, n) at the end of the last iteration i, and the increment of displacements Δu_n for the current iteration for the two nodes as $(u_n^i)_p$, $(\Delta u_n)_p$; $(u_n^i)_q$, $(\Delta u_n)_q$.

(2) Compute

$$A = [(u_n^i)_q + (\Delta u_n)_q + t - \{(u_n^i)_p + (\Delta u_n)_p\}] \tag{15}$$

(3) If $A \geq 0$ penetration does not occur and proceed to next pair of nodes, that is, go to step 1.

(4) If $A < 0$ penetration occurs, then compute

$$\alpha = \frac{[\varepsilon t - t - \{(u_n^i)_q - (u_n^i)_p\}]}{[(\Delta u_n)_q - (\Delta u_n)_p]} \tag{16}$$

Here εt = chosen value of thickness to be restored.

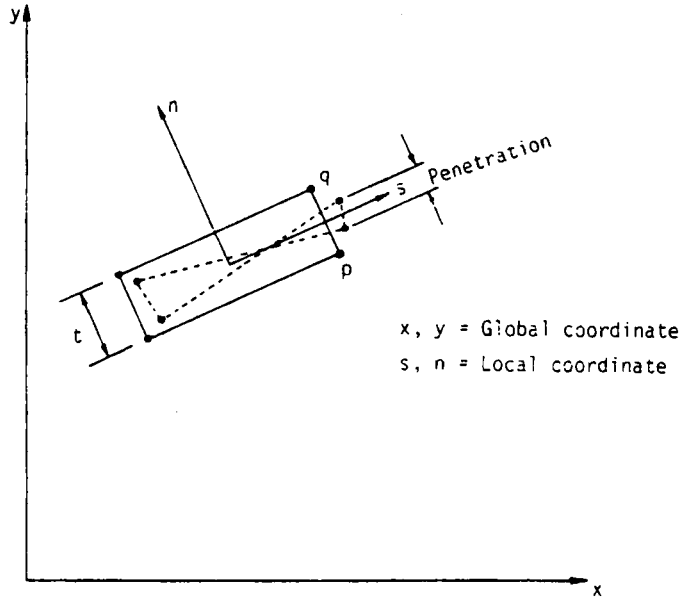

Figure 17. Schematic diagram of penetration of nodes

(5) Evaluate the amount by which the computed penetration is suppressed in order to establish thickness separation $= \varepsilon t$. For this purpose find

$$(\alpha - 1)(\Delta u_n)_p, (\alpha - 1)(\Delta u_n)_q \tag{17}$$

and transform them into global displacements

$$(\Delta u^1)_p, (\Delta v^1)_p; (\Delta u^1)_q, (\Delta v^1)_q$$

where u^1 and v^1 denote x and y components of global displacements. The vector of the suppressed global displacements is given by

$$\{\Delta q\}^T = [(\Delta u^1)_p, (\Delta v)^1_p, (\Delta u^1)_q, (\Delta v^1)_q] \tag{18}$$

Find the equivalent or correction load vector $\{\Delta Q\}$ as

$$\{\Delta Q\} = \int_V [C]_i[B]\{\Delta q\}\, dV \tag{19}$$

(6) Repeat the above steps for all pairs of interface nodes and accumulate the correction load vectors for all nodes that have experienced penetration.

(7) Perform finite element iterative analysis with the accumulated correction load until convergence, defined by

$$|A - \varepsilon t| \leq \lambda \tag{20}$$

where λ is a small number.

REFERENCES

1. R. E. Goodman, R. L. Taylor and T. L. Brekke, 'A model for the mechanics of jointed rock', *J. Soil Mech. and Found. Div. ASCE*, **94**, (SM 3) (1968).

2. O. C. Zienkiewicz *et al.*, 'Analysis of nonlinear problems with particular reference to jointed rock systems', *Proc. 2nd Intl. Conf. Society of Rock Mech.*, Belgrade, Vol. 3, 1970, pp. 501–509.
3. J. Ghaboussi, E. L. Wilson and J. Isenberg, 'Finite element for rock joints and interfaces', *J. Soil Mech. and Found. Div., ASCE*, **99**, (SM 10) (1973).
4. M. G. Katona *et al.*, 'CANDE—a modern approach for the structural design and analysis of buried culverts', *Report No. FltwA-RD. 77-J*, Federal Highway Administration, Washington, D.C., 1976.
5. M. G. Katona, 'A simple contact-friction interface element with applications to buried culvert', in C. S. Desai and S. K. Saxena (Eds), *Proc., Symp. on Impl. of Computer Procedures and Stress–Strain Laws in Geotech. Eng.*, Chicago, Illinois, Vol. 1, 1981, pp. 45–63.
6. L. R. Herrmann, 'Finite element analysis of contact problems', *J. Eng. Mech. Div., ASCE*, **104**, (EM 5), 1043–1057 (1978).
7. C. S. Desai, 'Soil–structure interaction and simulation problems', Chap. 7 in G. Gudehus (Ed.), *Finite Elements in Geomechanics*, Wiley, Chichester, 1977.
8. E. L. Wilson, 'Finite elements for foundations, joints and fluids', Chap. 10 in G. Gudehus (Ed.), *Finite Elements in Geomechanics*, Wiley, Chichester, 1977.
9. C. S. Desai and J. T. Christian (Eds), *Numerical Methods in Geotechnical Engineering*, McGraw-Hill, New York, 1977.
10. C. S. Desai, 'Some aspects of constitutive laws of geologic media', in W. Wittke (Ed.), *Proc. 3rd Int. Conf. on Num. Methods in Geomech.*, Aachen, W. Germany, 1979.
11. C. S. Desai, 'Behavior of interfaces between structural and geologic media', *Proc. Intl. Conf. on Recent Advances in Geotech. Earthquake Eng. and Soil Dynamics*, St. Louis, Mo., 1981.
12. J. G. Lightner and C. S. Desai, 'Improved numerical procedures for soil–structure interaction including simulation of construction sequences', *Report No. VPI-E-79.32*, Dept. of Civil Eng., Va. Tech., Blacksburg, Va., 1979.
13. J. G. Lightner, 'A mixed finite element procedure for soil–structure interaction including construction sequences', *Ph.D. Dissertation*, Dept. of Civil Eng., Virginia Tech., Blacksburg, Va., 1981.
14. H. J. Siriwardance, 'Nonlinear soil–structure interaction analysis of one-, two-, and three-dimensional problems using finite element method', *Ph.D. Dissertation*, Dept. of Civil Eng., Virginia Tech, Blacksburg, Va., 1980.
15. S. M. Sargand, 'A hybrid finite element procedure for soil–structure interaction including construction sequences', *Ph.D. Dissertation*, Dept. of Civil Eng., Virginia Tech, Blacksburg, Va, 1981.
16. M. M. Zaman, 'Influence of interface behaviour in dynamic soil–structure interaction problems', *Ph.D. Thesis*, Dept. of Civil Eng. and Eng. Mech., Univ. of Arizona, Tucson, AZ, 1982.
17. C. S. Desai, M. M. Zaman and E. C. Drumm, 'Testing and modelling for interfaces under cyclic loading', under preparation.
18. G. N. Pande and K. G. Sharma, 'On joint/interface elements and associated problems of numerical ill-conditioning', *Short Comm., Int. j. numer. anal. methods geomech.*, **3**, 293–300 (1979).
19. J. Isenberg and D. K. Vaughan, 'Nonlinear effects in soil–structure interaction', in C. S. Desai and S. K. Saxena (Eds), *Proc., Symposium on Implem. of Computer Procedures and Stress–Strain Laws in Geotech. Eng.*, Chicago, Illinois, Vol. 1, 1981, pp. 29–44.
20. C. S. Desai, J. G. Lightner and S. Sargand, 'Mixed and hybrid procedures for nonlinear problems in geomechanics', *Proc. 4th Int. Conf. Num. Methods in Geomech.*, Univ. of Alberta, Edmonton, Canada, 1982.
21. G. W. Clough and J. M. Duncan, 'Finite element analyses of retaining wall behavior', *J. Soil Mech. and Found. Div., ASCE*, **97**, (SM 12) (1971).
22. C. S. Desai, 'Numerical design-analysis of piles in sands', *J. Geotech. Eng. Div., ASCE*, **100**, (GT 6), 613–625 (1974).
23. C. S. Desai, 'Finite element method for analysis and design of piles', *Misc. Paper S-76-21*, U.S. Army Engr. Waterways Expt. Stn., Vicksburg, Mississippi, Oct. 1976.
24. C. S. Desai, 'Cyclic multi-degree-of-freedom shear device', *Report*, Dept. of Civil Eng., Virginia Tech, Blacksburg, Va, 1980.
25. J. W. Burns and R. M. Richards, 'Attenuation of stresses for buried cylinders', *Proc., Symp. on Soil–Structure Interaction*, Univ. of Arizona, Tucson, AZ, 1964.
26. C. J. F. Clausen, 'Finite element analysis of the strutted excavation at Vaterland 1', *Reports No. 52601-2*, Norwegian Geotechnical Institute, Oslo, Norway, 1971.
27. A. I. Mana, 'Finite element analyses of deep excavation behavior in soft clay', *Ph.D. Dissertation*, Stanford Univ., Stanford, Calif., 1978.
28. C. S. Desai, L. D. Johnson and C. M. Harget, 'Analysis of pile-supported gravity lock', *J. Geotech. Eng. Div., ASCE*, **100**, (GT 9) (1974).